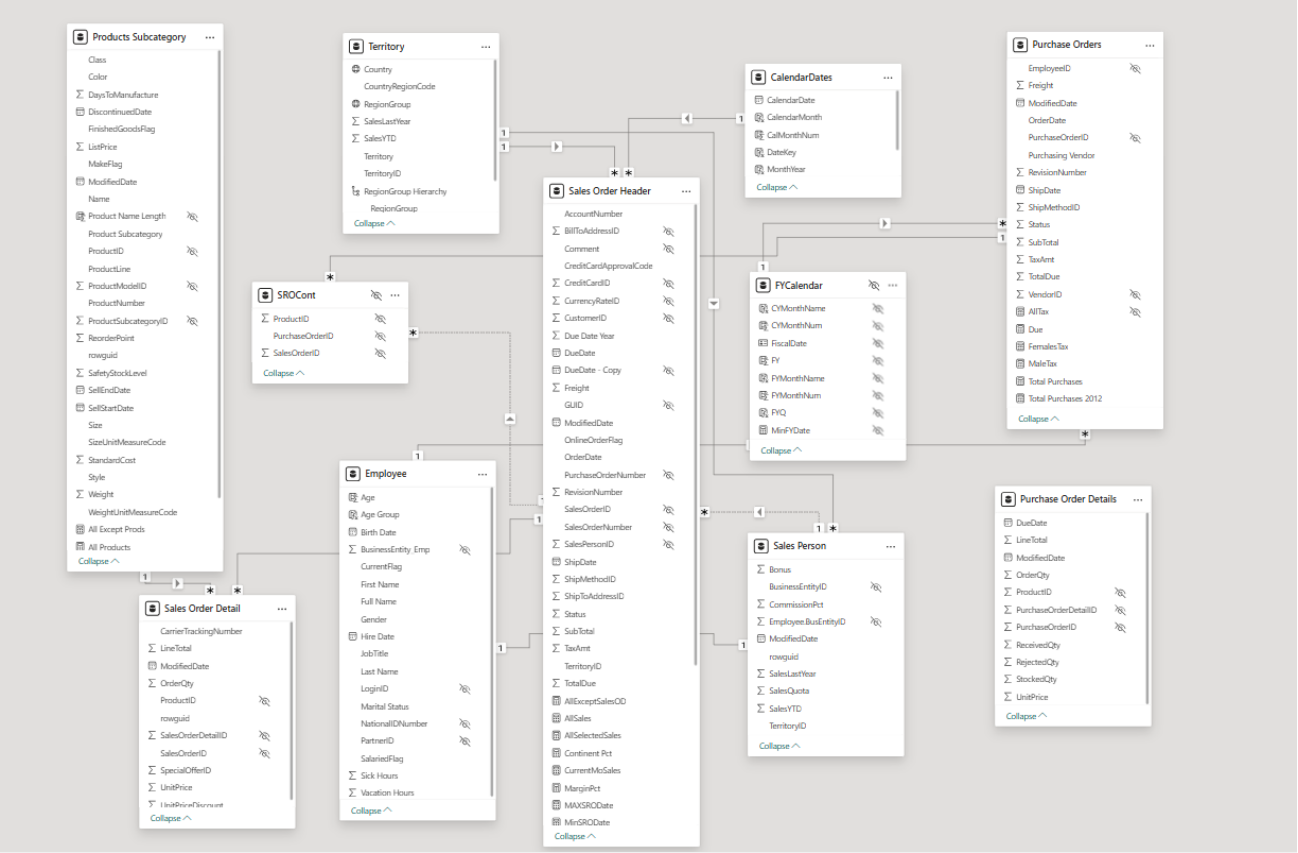
Power BI Sales Purchase Details and HR report

This report demonstrates an end-to-end Power BI build that combines **Sales Orders**, **Purchase Orders**, **Product Standard Cost**, **Territory profitability**, and **HR analytics** into a single semantic model and a set of interactive report pages.

Primary audience: Sales operations, procurement, finance/FP&A, and business leadership who need a unified view of revenue, cost, and operational drivers.

Data model:

Purpose: Document the semantic model behind the report and how the subject areas (Sales, Purchasing, Cost, Territory, and HR) are connected to enable consistent, cross-page analysis.
Data source (AdventureWorks 2022): The dataset for this project is imported from the **AdventureWorks 2022** sample database in **Microsoft SQL Server**. The schemas used for the data model are Sales (Orders & customers), Purchasing (Procurement), Products & Cost, and HR.
Model design: Star-schema approach with shared dimensions (Date, Product, Customer/Territory, Vendor/Employee as applicable) connected to multiple fact tables (Sales Orders, Purchase Orders, Standard Cost, and HR facts).
Reusability: Centralized measures ensure KPIs (Revenue, Cost, Margin, Headcount, etc.) remain consistent across pages and visuals.



Data Tables, Measures and relationships:

This section highlights the **AdventureWorks 2022 (SQL Server)** tables loaded into the model, the **DAX measure layer** used to calculate KPIs, and the **relationships** that enable slicing the same metrics by product, time, geography/territory, and organizational attributes. Using AdventureWorks also demonstrates working with a realistic, multi-schema database similar to what you would see in an enterprise environment.

- **Core KPIs:** Total Sales, Total Purchases, Standard Cost, Gross Profit, Gross Margin %.
- **Time intelligence:** YTD/MTD, rolling periods, prior period comparisons using a dedicated Date table.
- **Quality patterns:** Use of DIVIDE for safe calculations, CALCULATE for context transition, and well-named measures for readability.
- **Skills demonstrated:** Measure-driven reporting (minimal calculated columns), consistent formatting, and a scalable semantic layer.

Measures (32)
- All Except Prods
- All Products
- All Selected Prods
- AllExceptSalesOD
- AllSales
- AllSelectedSales
- AllTax
- Continent Pct
- CurrentMoSales
- Due
- FemalesTax
- MaleTax
- MarginPct
- MAXSRODate
- MinFYDate
- MinSRODate
- Number Of Sales
- ParallelMonthSales
- ParallelQtSales
- ParallelYearSales
- Previous Sales MTD
- Previous YTD
- SalesMTD
- SalesProfit
- SalesYTD
- Total Product Cost
- Total Products
- Total Purchases
- Total Purchases 2012
- TotalCost
- TotalSales
- YOY Sales

Relationships:

Relationships are configured so that shared dimensions (especially **Date**, **Product**, and **Territory**) filter each fact table consistently. This enables cross-highlighting between visuals and supports drill-through pages without breaking context.

Relationships (11)
- Purchase Orders[EmployeeID] <— Employee[PartnerID]
- Purchase Orders[OrderDate] <— FYCalendar[FiscalDate]
- Sales and Purchase Orders[PurchaseOrderID] <— Purchase Orders...
- Sales and Purchase Orders[SalesOrderID] <— Sales Order Header[...
- Sales Order Detail[ProductID] <— Products Subcategory[ProductID]
- Sales Order Detail[SalesOrderID] <— Sales Order Header[SalesOrd...
- Sales Order Header[OrderDate] <— CalendarDates[CalendarDate]
- Sales Order Header[SalesPersonID] <— Sales Person[BusinessEntit...
- Sales Order Header[TerritoryID] <— Territory[TerritoryID]
- Sales Person[BusinessEntityID] — Employee[PartnerID]
- Sales Person[TerritoryID] <— Territory[TerritoryID]

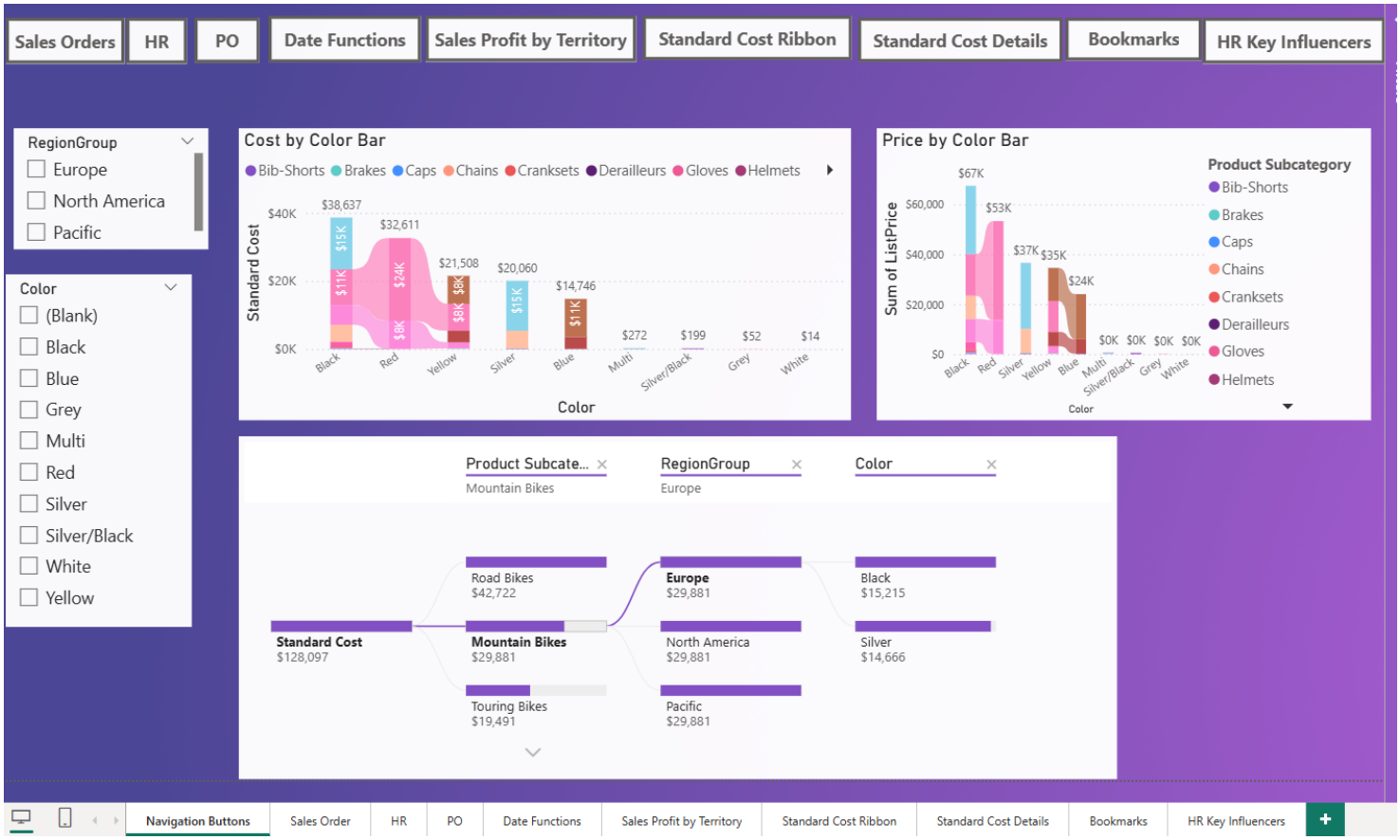
Power BI Report Pages:

The pages below demonstrate report design and end-user experience features such as **navigation**, **interactive filtering**, **drill-through**, **bookmarks**, **decomposition/AI visuals**, and **time-based analysis**. Each page is described in terms of business questions answered and Power BI skills showcased.

Navigation Page:

Purpose: Provide a landing page that makes the report easy to explore, guiding users to the right subject area (Sales, Purchasing, Cost, Territory, HR) with consistent navigation experience.

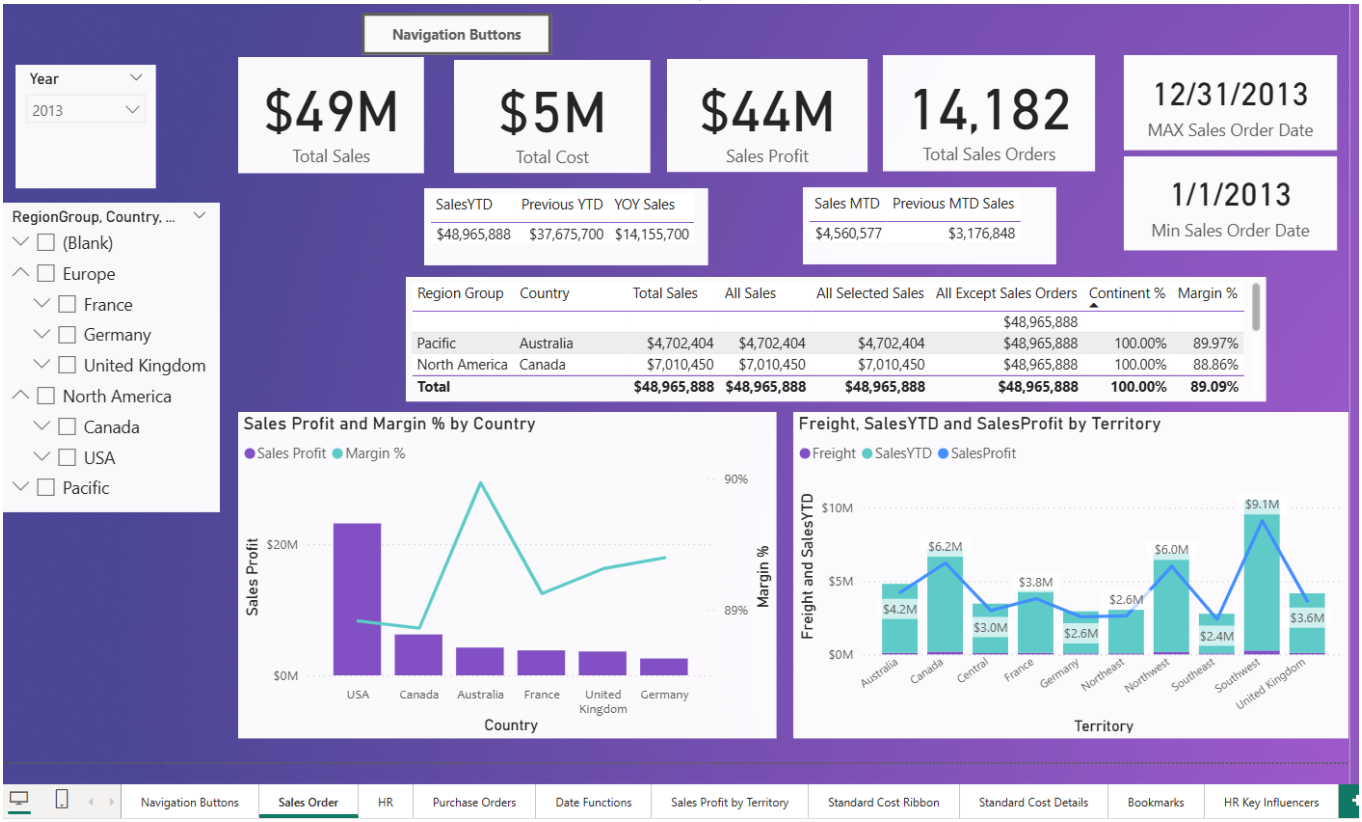
- **Usability features:** Clear page grouping, intuitive labels, and a "start here" experience for first-time users.
- **Navigation techniques:** Buttons/page navigation and/or bookmark-driven navigation (depending on the implementation in the PBIX).
- **Skills demonstrated:** Report layout design, consistent theme usage (colors/fonts), and user-centric information architecture.
- **Visuals on the page:** Page navigation buttons, supporting images/icons, and optional slicers/instructions for how to use the report.



Sales Order:

Purpose: Track sales order performance and customer demand patterns, with the ability to slice results by date, product, customer, and territory to identify revenue drivers and areas needing attention.

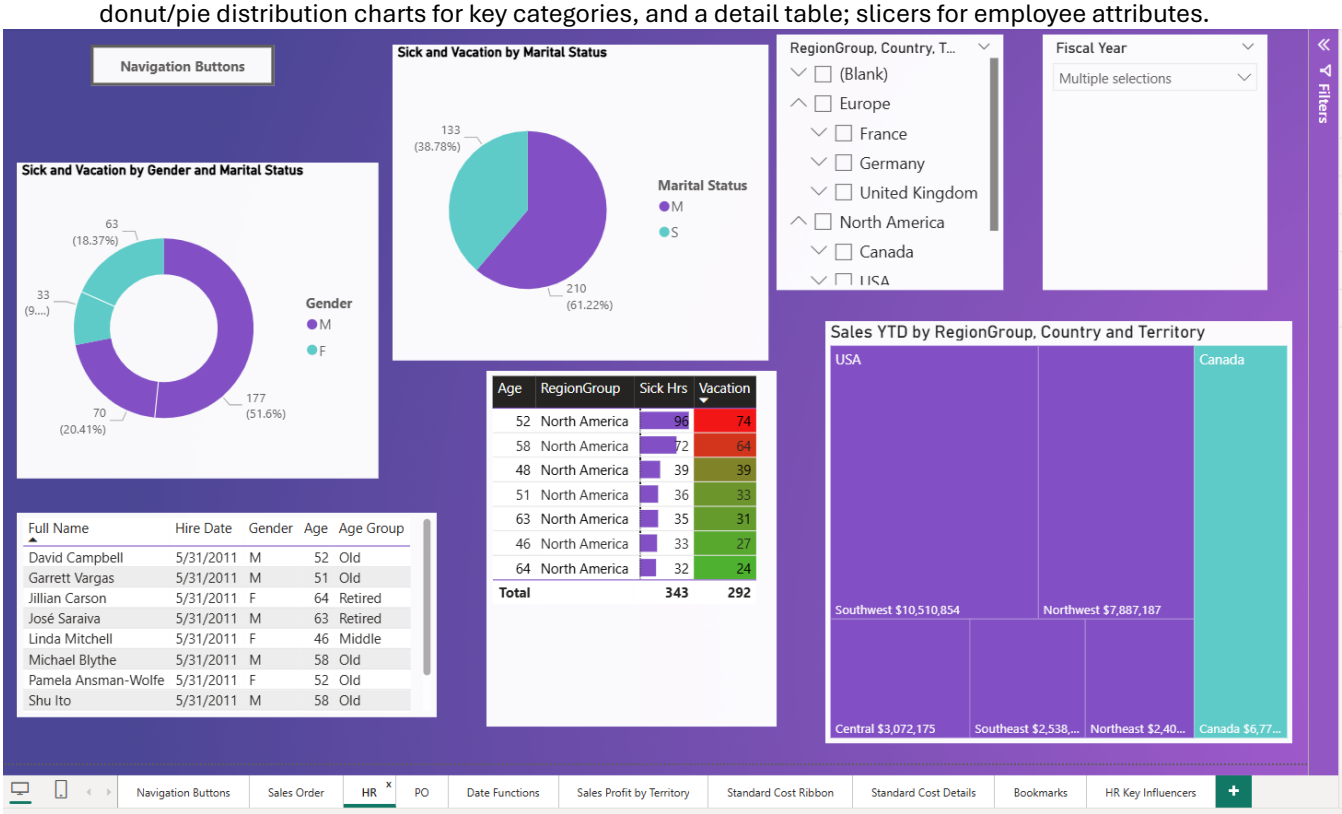
- **Key questions:** Which products/territories drive the most sales? How do sales trend over time? Which customers contribute most to revenue?
- **Interactivity:** Cross-filtering across charts and tables; slicers for time period and business dimensions; drill-down where hierarchies exist (e.g., Year → Quarter → Month).
- **Skills demonstrated:** KPI cards, trend visuals, top-N analysis, and measure-driven totals that respond to filter context.
- **Visuals on the page:** KPI cards, line/area trend chart, bar/column chart for top products or customers, and a detail table/matrix with sales order breakdown; slicers for Date and business dimensions.



HR (Human Resources):

Purpose: Provide an HR overview of workforce composition and key people metrics (for example: headcount, department distribution, demographics, and attrition-related indicators) to support staffing and retention decisions.

- **Key questions:** What is the current headcount and how is it distributed by department/job role? Are there patterns by tenure or other employee attributes?
- **Analysis approach:** Dimension-driven slicing (Department, Job Title, Education, Marital Status, etc.) with measures for counts and rates.
- **Skills demonstrated:** Clean categorical visuals, consistent measure logic, and preparing the model to support AI visuals (Key Influencers / Top Segments).
- **Visuals on the page:** KPI cards (e.g., headcount), bar/column charts by department/job role, donut/pie distribution charts for key categories, and a detail table; slicers for employee attributes.



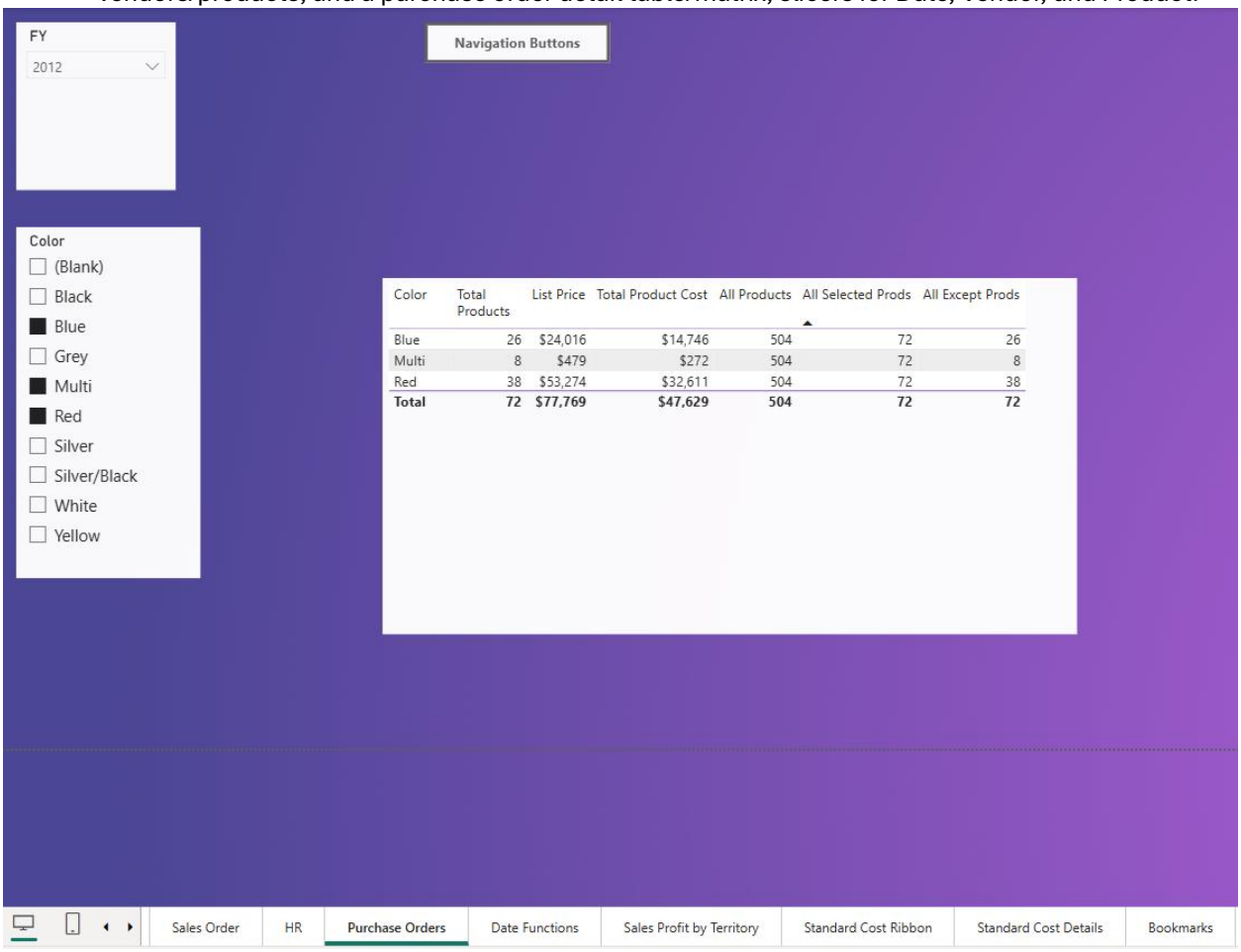
Full Name	Hire Date	Gender	Age	Age Group
David Campbell	5/31/2011	M	52	Old
Garrett Vargas	5/31/2011	M	51	Old
Jillian Carson	5/31/2011	F	64	Retired
José Saraiva	5/31/2011	M	63	Retired
Linda Mitchell	5/31/2011	F	46	Middle
Michael Blythe	5/31/2011	M	58	Old
Pamela Ansman-Wolfe	5/31/2011	F	52	Old
Shu Ito	5/31/2011	M	58	Old

Age	RegionGroup	Sick Hrs	Vacation
52	North America	96	74
58	North America	72	64
48	North America	39	39
51	North America	36	33
63	North America	35	31
46	North America	33	27
64	North America	32	24
Total		343	292

Purchase Orders:

Purpose: Monitor purchasing activity and supplier spend, enabling procurement and finance users to track order volume, purchased quantities, and cost trends by vendor, product, and time period.

- **Key questions:** Which vendors account for the most spend? What categories/products are being purchased most? Are purchase costs increasing over time?
- **Controls:** Date and vendor/product slicers to isolate a supplier or category; drill-down to understand spend drivers.
- **Skills demonstrated:** Spend analytics design, vendor-level segmentation, and consistent currency/number formatting.
- **Visuals on the page:** KPI cards (total spend/orders), line chart for spend trend, bar chart for top vendors/products, and a purchase order detail table/matrix; slicers for Date, Vendor, and Product.

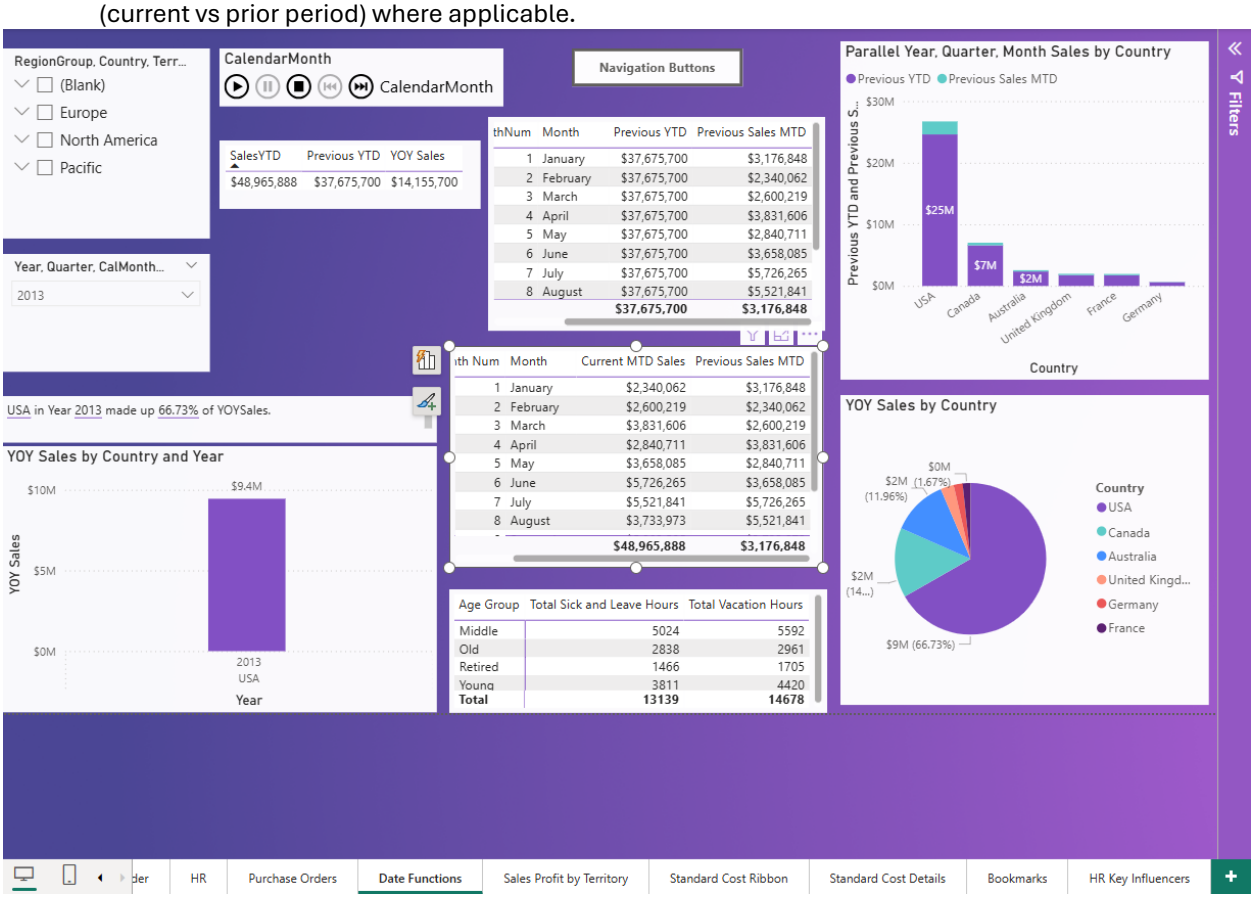


Color	Total Products	List Price	Total Product Cost	All Products	All Selected Prods	All Except Prods
Blue	26	$24,016	$14,746	504	72	26
Multi	8	$479	$272	504	72	8
Red	38	$53,274	$32,611	504	72	38
Total	**72**	**$77,769**	**$47,629**	**504**	**72**	**72**

Date Functions in Power BI:

Purpose: Showcase DAX time-intelligence patterns and a robust Date table that supports consistent period-to-date and prior-period comparisons across Sales and Purchasing pages. The PlayAxis Visual automatically displays the other visual changes over calendar months as it plays.

- **Date table fields:** Year, Quarter, Month, Month Name, Week, Day, and sortable columns for correct chronology.
- **Time-intelligence measures:** Sales YTD, Sales MTD, Prior Year Sales, YoY %, Rolling 12 Months (examples).
- **Skills demonstrated:** Marking a Date table, building reusable time measures, and ensuring visuals aggregate correctly at different time grains.
- **Visuals on the page:** Time-series visuals (line/column), and a **Play Axis** (animated time control) that plays through months while updating the associated chart(s); Date slicers and comparison KPI cards (current vs prior period) where applicable.



Sales Profit by Territory:

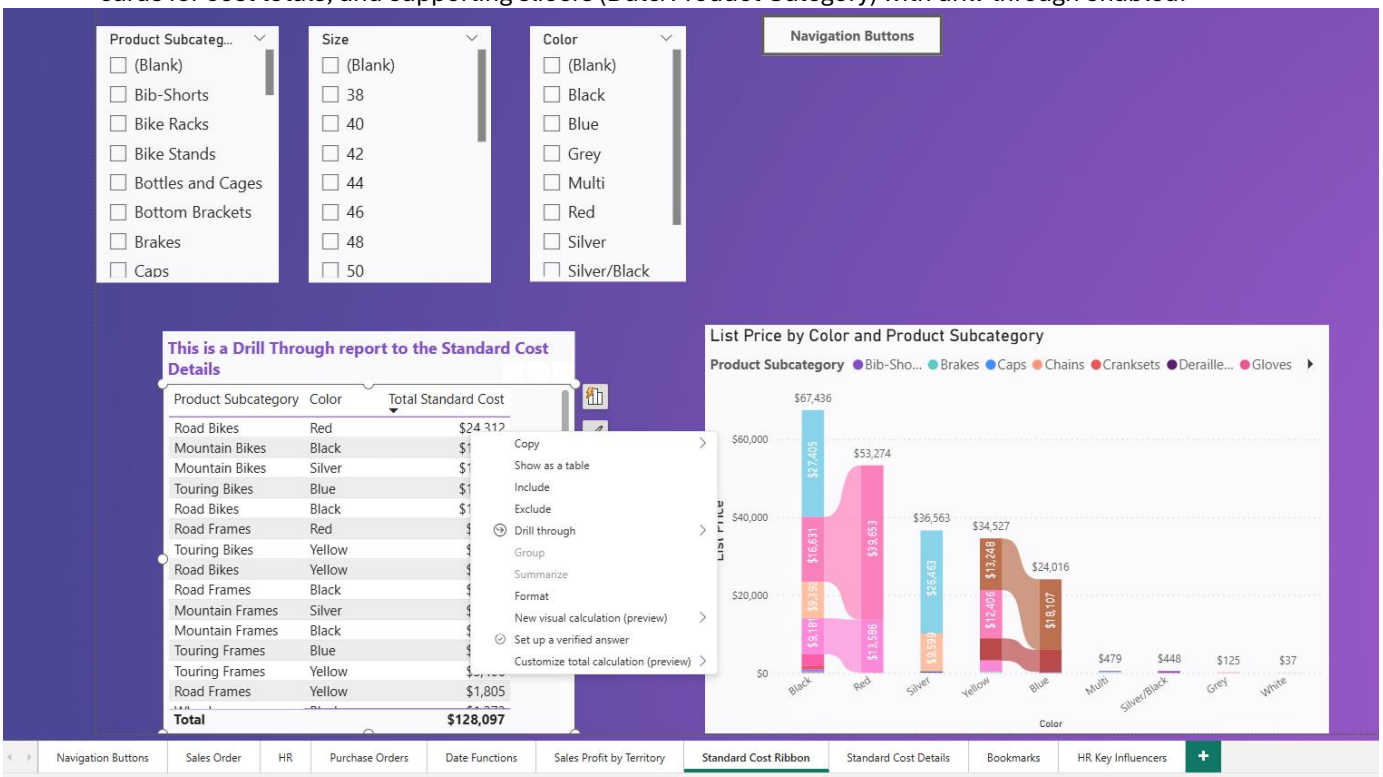
Purpose: Analyze profitability by geography/territory to identify where the business is earning the highest profit and margin, and to spot territories where cost-to-serve or discounting may be reducing profitability.

- **Metrics:** Sales, Cost, Profit, Margin %, and contribution by territory.
- **Interactivity:** Click a territory to filter supporting visuals (products/customers); drill-down from region → sub-region where applicable.
- **Skills demonstrated:** Profit modeling with DAX, geographic segmentation, and designing visuals that tell a margin story (not just revenue).
- **Visuals on the page:** Map (or filled map) and/or bar charts by territory, KPI cards for profit/margin, and supporting charts/tables for products/customers filtered by the selected territory.

Standard Cost Summary Ribbon Report:

Purpose: Provide a high-level view of **standard cost** by product/category and highlight cost drivers at a glance. The ribbon-style visual helps compare rankings or changes across periods and supports quick "what changed" analysis.

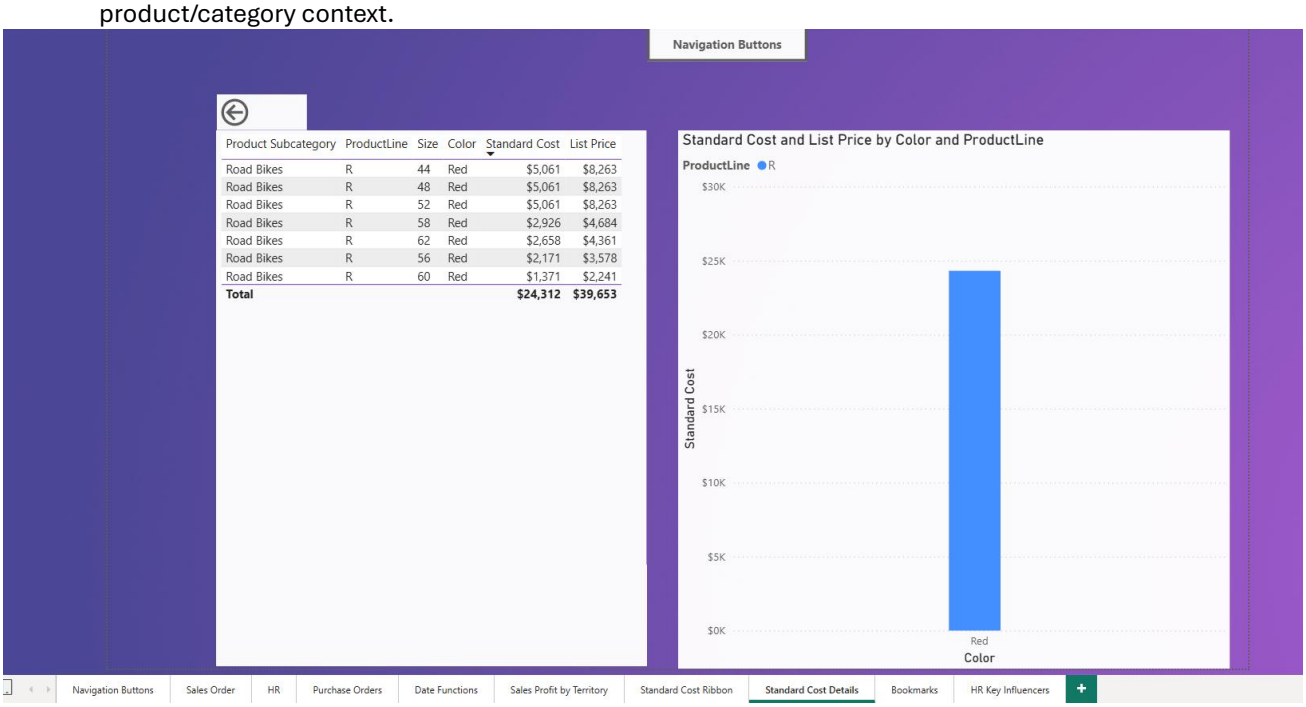
- **Use case:** Identify products with the highest standard cost and compare their position over time or across categories.
- **Navigation:** Designed as a **drill-through entry point** to jump into product-level cost detail.
- **Skills demonstrated:** Visual selection for the business question, consistent measure reuse, and configuring drill-through filters.
- **Visuals on the page:** Ribbon chart for ranking standard cost by product/category across time, KPI cards for cost totals, and supporting slicers (Date/Product Category) with drill-through enabled.



Standard Cost Details with Drill through from Standard Cost Ribbon:

Purpose: Provide detailed, record-level or product-level breakdown of standard cost after a user drill through from the summary ribbon. This page supports root-cause analysis by showing which components, dates, or categories explain high cost.

- **Drill-through:** Carries the selected Product/Category context from the summary page, so users do not need to re-filter.
- **Detail analysis:** Tables/matrices for cost components and trends; sorting and filtering to isolate the main drivers.
- **Skills demonstrated:** Drill-through configuration, detail page design, and maintaining consistent context across navigation.
- **Visuals on the page:** Drill-through filter panel (implicit), detail table/matrix for product cost breakdown, trend chart for standard cost over time, and KPI cards summarizing the selected product/category context.

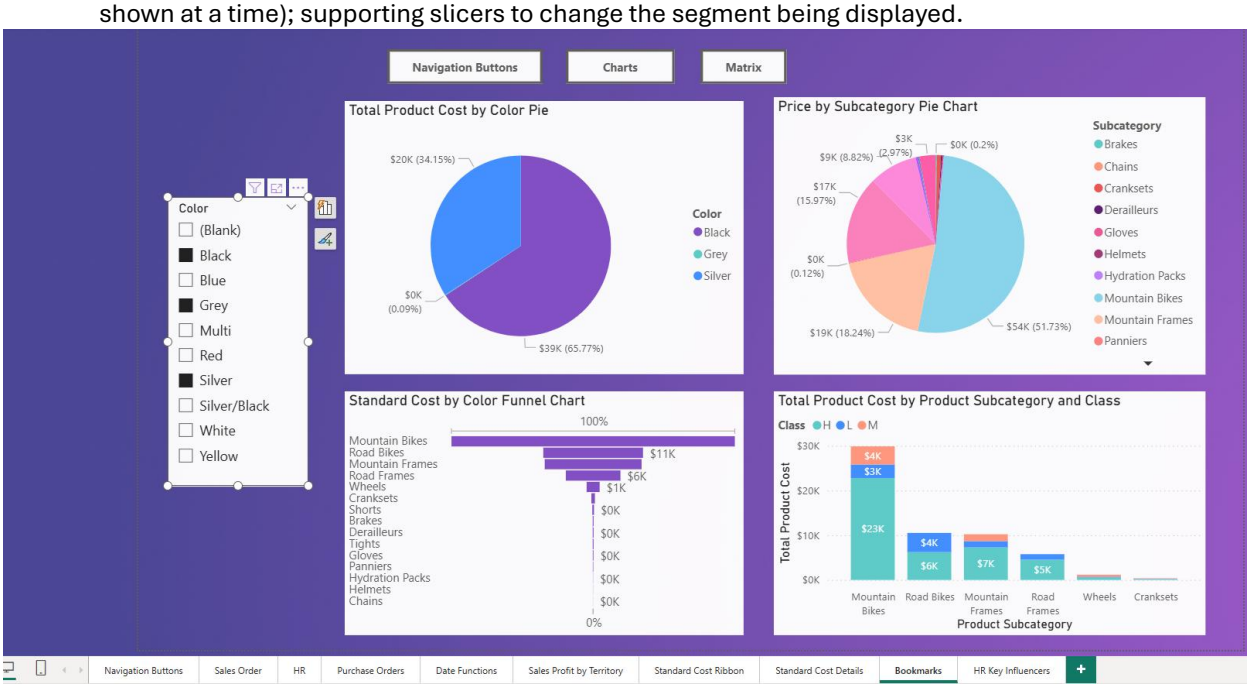


Product Subcategory	ProductLine	Size	Color	Standard Cost	List Price
Road Bikes	R	44	Red	$5,061	$8,263
Road Bikes	R	48	Red	$5,061	$8,263
Road Bikes	R	52	Red	$5,061	$8,263
Road Bikes	R	58	Red	$2,926	$4,684
Road Bikes	R	62	Red	$2,658	$4,361
Road Bikes	R	56	Red	$2,171	$3,578
Road Bikes	R	60	Red	$1,371	$2,241
Total				**$24,312**	**$39,653**

Bookmark Pie and Funnel Charts:

Purpose: Demonstrate **bookmarks** and **selection-based storytelling** by letting the user toggle between different visual perspectives (for example, distribution view via pie chart vs. process/conversion view via funnel chart) without leaving the page.

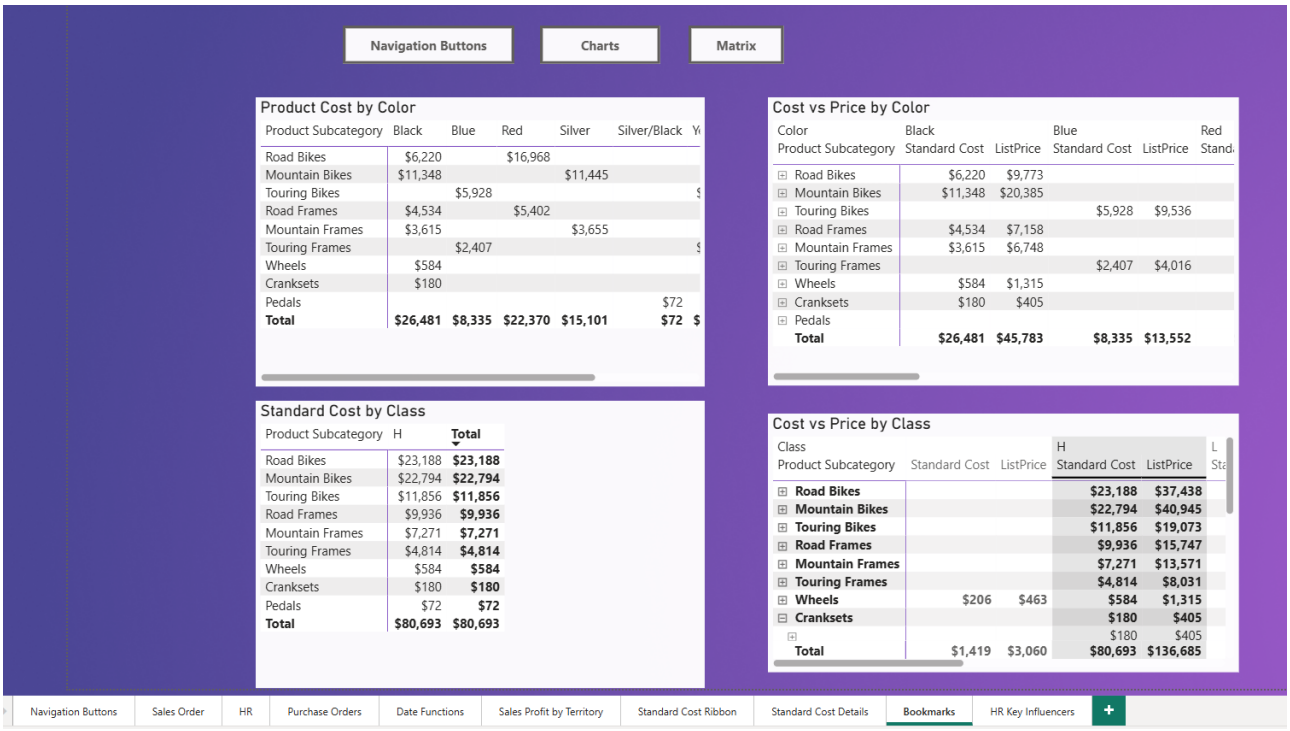
- **Bookmark behavior:** Uses the Selection pane to show/hide visuals and preserve filters/slicers as needed.
- **User experience:** Buttons act like tabs to switch the view; reduces clutter while still offering multiple analyses.
- **Skills demonstrated:** Bookmark management, button configuration, and designing multi-view pages with a consistent layout.
- **Visuals on the page:** Bookmark toggle buttons plus a **pie chart** view and a **funnel chart** view (only one shown at a time); supporting slicers to change the segment being displayed.



Bookmark Matrix:

Purpose: Showcase a bookmark-driven **matrix/table view** that can switch between different breakdowns (e.g., by product, by territory, by vendor) while keeping the page layout stable for the user.

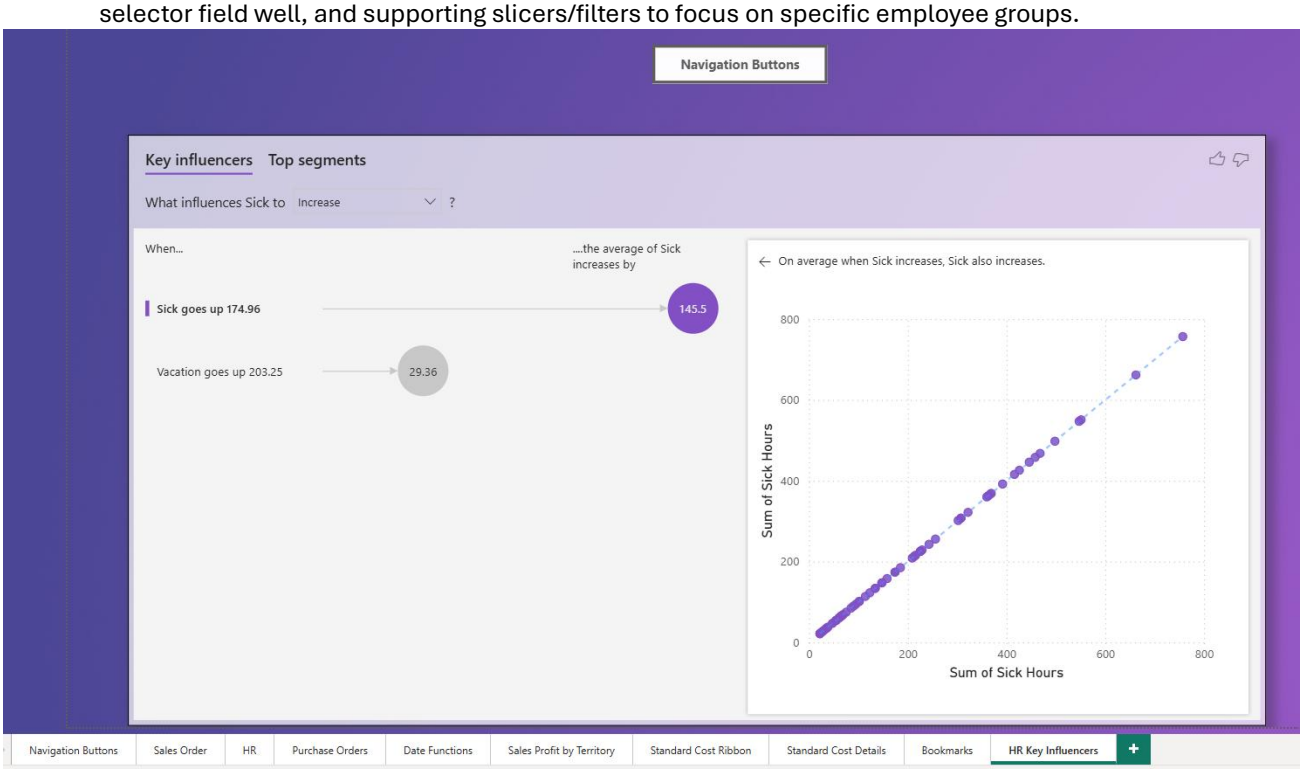
- **Matrix features:** Hierarchical drill (expand/collapse), subtotals, and sorting to quickly identify top contributors.
- **Dynamic analysis:** Bookmarks enable "view switching" without duplicating pages, supporting multiple stakeholder needs.
- **Skills demonstrated:** Matrix formatting, conditional emphasis where applicable, and bookmark/navigation design patterns.
- **Visuals on the page:** Bookmark buttons (to switch the matrix view) and a **matrix** visual with hierarchical rows/columns, subtotals, sorting, and optional conditional formatting; slicers to filter the matrix.



HR Key Influencers:

Purpose: Use Power BI's **Key Influencers** AI visual to identify which employee attributes (such as job level, overtime, tenure, department, etc.) most strongly relate to a selected outcome (commonly attrition, performance rating, or satisfaction depending on the dataset).

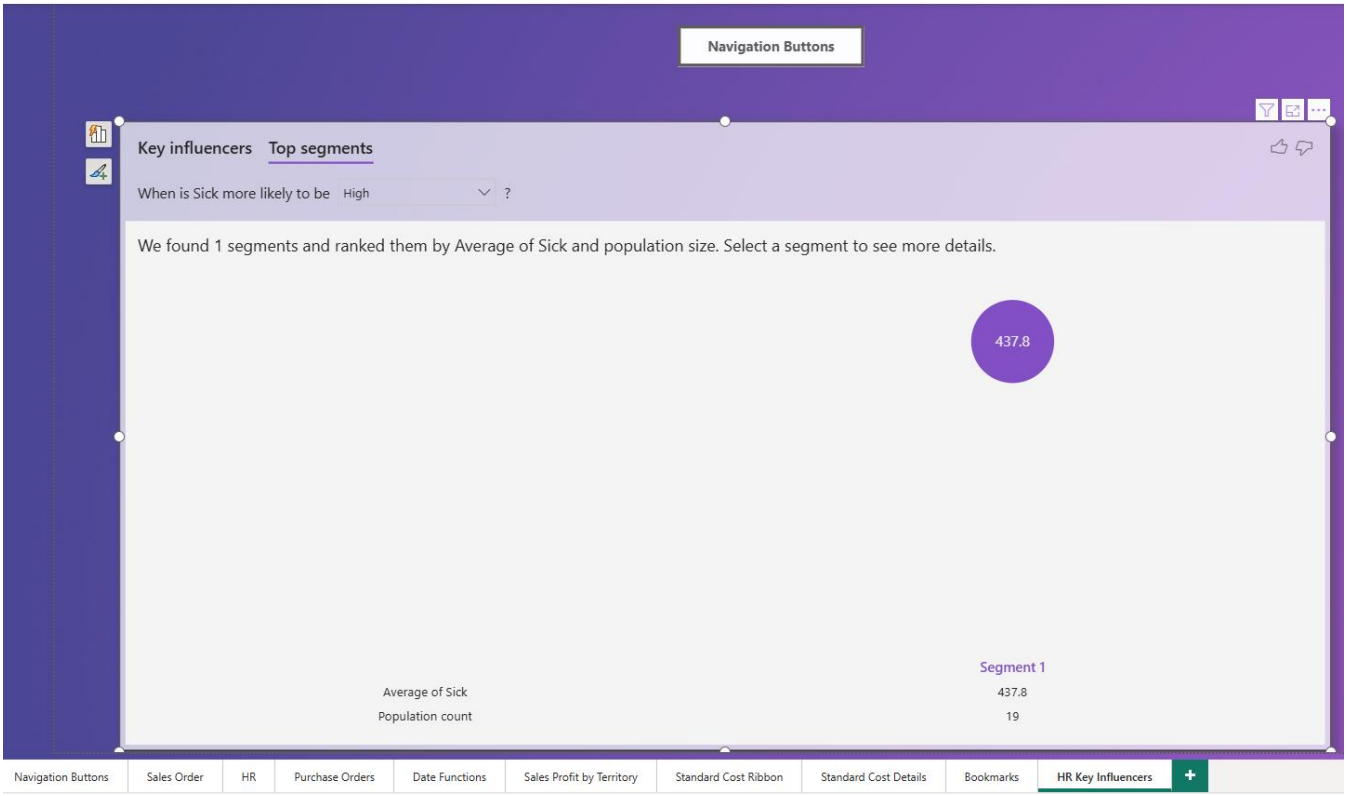
- **How to read it:** Select the outcome and review top drivers ranked by influence; click drivers to filter and explore supporting visuals.
- **Business value:** Helps HR focus on the factors most associated with risk/high outcomes for targeted interventions.
- **Skills demonstrated:** Preparing categorical/numeric fields for AI visuals, ensuring clean data types, and building explainable analytics pages.
- **Visuals on the page: Key Influencers** AI visual (influencer list and explanation pane), outcome selector field well, and supporting slicers/filters to focus on specific employee groups.



HR Key Influencer, Top Segments – High and Low:

Purpose: Show the **Top Segments (High/Low)** results from the Key Influencers analysis—groups of employees that share characteristics associated with a higher/lower likelihood of the selected outcome.
Interpretation: Each segment combines multiple attributes; the size and lift indicate how strongly it differs from the overall population.

- **Actionability:** Supports targeted retention/training plans by focusing on the most impacted employee groups.
- **Skills demonstrated:** AI visual configuration, segment interpretation, and connecting insights back to business decisions.
- **Visuals on the page: Key Influencers** visual set to **Top segments** (segment cards with size and lift), plus slicers/filters to explore the segment definitions and compare populations.



Key Insights (from visuals):

- **Navigation Page:** Designed to reduce user friction—users can quickly jump to Sales, Purchasing, Cost, Territory, or HR analysis without needing to understand the model structure.
- **Sales Order:** Reveals sales concentration (top products/customers drive a large share of revenue) and makes it easy to spot trend changes over time (seasonality or growth/decline) by interacting with Date slicers and trend visuals.
- **HR (Overview):** Highlights how headcount and workforce composition vary by department/job role and helps identify segments that differ from the overall workforce distribution using slicers and categorical visuals.
- **Purchase Orders:** Identifies the vendors and products/categories that contribute most to spend and shows procurement trends over time—useful for supplier rationalization and budgeting conversations.
- **Date Functions (Play Axis):** Demonstrates how the business changes month-over-month by animating the timeline—useful for communicating trends and highlighting periods where measures shift.
- **Sales Profit by Territory:** Shows that the highest-revenue territories are not always the highest-margin territories—profitability analysis helps prioritize where to grow vs. where to fix margin leakage.
- **Standard Cost Summary (Ribbon):** Quickly surfaces which products/categories rank highest in standard cost and whether rankings shift by time period—supports fast identification of cost drivers to investigate.
- **Standard Cost Details (Drill-through):** Enables root-cause analysis by taking the selected product/category into a detail view so users can validate which components/time periods explain high standard cost.
- **Bookmarks (Pie/Funnel):** Supports two complementary views of the same problem (distribution vs. stage-based view) without duplicating pages, helping users compare perspectives quickly.
- **Bookmarks (Matrix):** Gives power users a detail-first, sortable view and allows switching the breakdown (e.g., product vs. territory vs. vendor) while keeping the layout consistent.
- **HR Key Influencers + Top Segments:** Identifies the strongest drivers associated with the selected HR outcome and summarizes employee segments that are most/least likely to experience that outcome, turning analysis into actionable targeting.